

		Fitting the Parametric models for survival data.		
	Sections from References:			
Books and References:				
1. Klein J.P. and Moeschberger M.L. (2003) Survival Analysis - Techniques for censored and truncated data, Second Edition, Springer-Verlag , New York. 2. Lawless J.F (2003) Statistical Models and Methods for Lifetime Data, Second Editon, John Wiley & Sons, Relevant Sections of the Chapters 9. 3. Kalbfleisch J.D and Prentice, R.L. (2002) The Statistical Analysis of Failure Time Data, Second Edition, John Wiley & Sons Inc. 4. Deshpande, J .V. and Purohit, S. G. (2006). Lifetime Data: Statistical Models and Methods. World Scientific.				

Programme	B. Sc. Statistics				
Course Code	STA6EJ304				
Course Title	Demography				
Type of Course	Major Elective				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites					
Course	On completion of the course, the students shall be able to Understand				

Summary Objective	basics of Statistical Techniques used in population data analysis.
-------------------	--

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand various sources of demographic data	U	C	Instructor-created exams / Quiz
CO2	Understand life tables and their main features	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Calculate and interpret mortality and fertility measures	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Analyze internal migration and its measurement, exploring migration models	U	C	Instructor-created exams / Home Assignments
CO5	Apply demographic concepts and measures practically using data analysis tools like R or Excel.	Ap	P	One Minute Reflection Writing assignments
CO6		Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I			10	15
	1	Sources of demographic data		
	2	Census and Registration		
	3	Ad-hoc surveys, Hospital records		
	4	Demographic profiles of the Indian Census.		

	Sections from References:			
II			10	15
	5	Complete life table and its main features		
	6	Uses of life table. Makehams and Gompertz curves.		
	7	National life tables. UN model life tables.		
	8	Abridged life tables. Stable and stationary populations.		
	Sections from References:			
III			16	20
	9	Measurement of Mortality: Crude death rate		
	10	Standardized death rates		
	11	Age-specific death rates		
	12	Infant Mortality rate		
	13	Death rate by cause		
	14	Measurement of Fertility: Crude birth rate		
	15	General fertility rate		
	16	Age specific birth rate		
	17	Total fertility rate		
	Sections from References:			
IV			12	20
	18	Gross reproduction rate, Net reproduction rate		
	19	Internal migration and its measurement, migration models		
	20	Concept of international migration		
	21	Net migration. International and postcensal estimates		
	22	Decennial population census in India		
	Sections from References:			
V	Open Ended Module:		12	
	1	Hands-on in R or Excel: Mortality and fertility measures.		
	Sections from References:			
Books and References:				